

GEI Consultants



Geotechnical Investigation Report

GRCA 318719: Rehabilitate North Rim Utility Infrastructure

Grand Canyon Inner Canyon, Arizona

Submitted to:
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August 11, 2025
Project No. 2305987





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VIA EMAIL: matt.gough@hdrinc.com

Matt Gough, P.E.
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Fort Collins, CO 80521

Re: Geotechnical Investigation Report
GRCA 318719: Rehabilitate North Rim Utility Infrastructure
Grand Canyon Inner Canyon, Arizona

Mr. Gough:

Submitted herewith is the Geotechnical Engineering Investigation for the Inner Canyon portion of the Grand Canyon National Park Rehabilitate North Rim Utility Infrastructure project. This study was conducted in accordance with our contract between GEI Consultants, Inc. and HDR Engineering, Inc. This report contains the results of our findings, an engineering interpretation with respect to the available project characteristics, and recommendations to aid design and construction of earth-related phases of this project.

If you have any questions regarding the contents of this report, please contact the undersigned.

Sincerely,

GEI Consultants, Inc.

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1. Introduction and Project Background

The National Park Service (NPS) is planning several upgrades to their existing water treatment and distribution infrastructure as part of the GRCA 318719: Rehabilitate North Rim Infrastructure Project. The purpose of the project is to rehabilitate existing drinking water supply infrastructure and includes improvements and upgrades to existing treatment facilities, replacement of the North Rim Waterline via directional drilling and installation of a cased bore from the North Rim to the Roaring Springs Pump House, and various other improvements to the inner canyon infrastructure. The NPS retained HDR Engineering, Inc. (HDR) as the lead engineer for this project. HDR retained GEI Consultants, Inc. (GEI) to perform various geotechnical investigations and design services for the various infrastructure improvements.

The focus of this report is to present the findings of our geotechnical and engineering geologic site reconnaissance, and to provide geotechnical recommendations for design and construction of the various improvements proposed within the inner canyon area. Geotechnical reports covering the proposed improvements on the North Rim and the proposed bore are provided under separate cover.

The proposed inner canyon improvements covered by this report are located at the existing Roaring Springs Pumphouse (RSPH) site, between the RSPH and Manzanita Ranger Station Site, and at the Cottonwood Campground as shown in Figure 1. More specifically, the proposed improvements covered by this report are listed below and grouped by site:

RSPH Site

1. Replacement of existing overhead electric power poles at four locations between the North Rim and the RSPH
2. Stabilization of the existing rip-rap slope behind the pumphouse structure
3. Construction of a new 5,000-gallon fire water tank outside of the southeast side of the pumphouse structure
4. Rehabilitation of the existing crib wall and replacement of portions of the existing concrete slab
5. Construction of a new pipe/pedestrian bridge across Roaring Creek near the existing pipe crossing location

RSPH Site to Manzanita Ranger Station Power Distribution

1. Addition of new 3-phase electrical line adjacent to the existing steel waterline alignment

Cottonwood Campground Site Power Distribution

1. Addition of a new solar shade structure near the existing storage building

2. Geologic Setting

The sites are located in Roaring Springs Canyon and along Bright Angel Canyon of the Grand Canyon. The Grand Canyon is, on average, 1 mile deep and up to 10 miles wide. The range in age of the geologic units exposed within the Grand Canyon is 1.7 billion to 200 million years old. Almost 40 major rock units have been identified in the canyon, however, only two of those geologic units are relevant to the proposed inner canyon improvements including the Tapeats Sandstone and the overlying Bright Angel Formation.

The Tapeats Sandstone is approximately 525 million years old and unconformably overlies Precambrian sedimentary rock locally. The large discrepancy in age between the Tapeats Sandstone and the Precambrian rocks is due to a regional unconformity, a period of uplift and erosion resulting in missing time between the two units. This feature, often referred to as “the Great Unconformity”, is visible throughout the Grand Canyon and in the project site. The Tapeats Sandstone is described as medium to coarse-grained, thin bedded, tan to purple-red sandstone and conglomerate.

The Bright Angel Formation is approximately 515 million years old and conformably overlies the Tapeats Sandstone. The Bright Angel Formation is primarily siltstone and shale but also contains harder interbedded sandstone layers. The shale primarily weathers to form slopes and is described as fine-grained, micaceous, ripple-laminated, fossiliferous, and green to purple-red. The lower portion of the Bright Angel Formation contains an interbedded slope-forming transition zone with the underlying Tapeats Sandstone.

Two major fault systems are present along the project area. The Bright Angel Fault trends northeast along Bright Angel Canyon near the Manzanita Ranger Station and Cottonwood Campground. This is a major fault through the canyon and is considered Quaternary active by the USGS, meaning that movement has occurred along the fault within the past 1.6 million years. Another fault system, the Roaring Springs Fault, intersects the Bright Angel Fault near the mouth of Roaring Springs Canyon and trends northwest along Roaring Springs Canyon near the RSPH. Geologic mapping by Huntoon et al. (1996) indicates the Roaring Springs Fault generally has a reverse sense of motion with the downthrown side to the southwest and 60 to 130 feet of vertical offset noted along the fault. The Roaring Springs fault is not included in the USGS Quaternary Fault and Fold Database of the United States, which indicates the duration since last movement along this fault is either more than 1.6 million years or the date of last movement is not well-established.

Erosional and depositional processes continue to occur in the canyon today, most notably on slope-forming units and along creeks at the canyon bottom. The processes are visible as alluvial (water deposited), and colluvial (gravity deposited) deposits.

3. Geotechnical Investigation

3.1. Field Investigation

GEI conducted a field investigation at proposed inner canyon infrastructure improvement locations between November 7, 2023 and November 9, 2023. GEI also visited the RSPH site specifically on November 17, 2020 to observe the existing site conditions. The investigation included visual rock and soil classification, hand excavation of test pits at the Cottonwood Campground site, and visual observations of existing infrastructure. Geologic conditions encountered during the investigation are discussed in Section 4. Our observations of existing infrastructure, recommendations for geotechnical design, and recommendations for construction methods are presented in Sections 6 and 7.

3.2. Geophysical Investigation

GEI subcontracted Collier Geophysics to perform a geophysical investigation using various geophysical methods to further characterize the subsurface conditions at the project site. GEI summarized the results and interpretation of the geophysical investigation in the North Rim Geophysical Study letter provided to HDR.

4. Subsurface Conditions

Primary materials visually observed during the field investigation include alluvium (water deposited), colluvium (gravity deposited), and Bright Angel Formation, and Tapeats Sandstone. Groundwater was not evaluated during the field investigation. Photos of the observed conditions are presented in Appendix A.

4.1. Alluvial Deposits

Alluvium was observed along Roaring Creek near the RSPH (Photos 1, 2 Appendix A) and along Bright Angel Creek near the Manzanita Ranger Station and near Cottonwood Campground. The alluvial deposits had variable gradations but were generally granular, typically classifying as sand or gravel and often with subangular to rounded cobble or boulder content. Typical observed alluvial deposit soil classifications in accordance with USCS criteria generally included silty sand (SM), clayey sand (SC), silty gravel (GM), and clayey gravel (GC).

GEI hand-excavated two test pits into alluvium near the proposed solar shade structure locations at the Cottonwood Campground to depths of 8 to 18 inches below ground surface as shown in Photos 3 and 4 in Appendix A. The soils encountered in Test Pits TP-1 and TP-2 consisted of loose, brown, dry, silty sand with gravel (SM), with gravel particles up to 3 inches in diameter (Photos 3, 4 Appendix A).

4.2. Colluvial Deposits

Colluvium was observed adjacent to and above the RSPH, and on slopes along the RSPH to Manzanita Ranger Station Power Distribution alignment (Photos 5, 6 Appendix A). Colluvium deposits are often referred to as talus and talus slopes and are comprised of rock debris that accumulate at the base of a cliff, chute, or slope. The colluvial deposits had variable gradations but were generally granular, typically classifying as gravel and often with angular cobble or boulder content. Typical observed colluvial deposit soil classifications in accordance with USCS criteria generally included silty gravel (GM), and clayey gravel (GC). The thickness of colluvium on bedrock slopes in the observed areas is highly variable and can be 1 foot or less up to 10 feet or more.

4.3. Bright Angel Formation

Shale, sandstone, and siltstone of the Bright Angel Formation was observed on the slopes above the RSPH (Photo 5, Appendix A). The observed bedrock was described as soft to hard, slightly to moderately weathered, very fine to fine-grained, tan, reddish brown, purple-red, and olive. Observed bed thicknesses ranged from approximately 0.25 inches to 12 inches and comprised interbedded shale, sandstone, and siltstone. The Bright Angel Formation is predominantly a slope-forming unit, but with limited cliff bands up to approximately 10 feet tall where sandstone beds occur.

4.4. Tapeats Sandstone

The Tapeats Sandstone was observed in the stream bed below the RSPH, and along portions of the RSPH to Manzanita Ranger Station Power Distribution alignment (Photos 7, 8 Appendix A). The unit is

predominantly sandstone with infrequent very thin shale beds, and was described as medium to hard, slightly to moderately weathered, very fine to coarse-grained, and tan, purple-red, and gray. Observed bed thicknesses ranged from approximately 1/2 inch to 8 feet. The Tapeats Sandstone is predominantly a cliff forming unit where observed along the Power Distribution Alignment.

4.5. Groundwater

Depth to groundwater was not investigated by GEI. However, based on observations of topography and drainage features such as creeks, a significant amount of groundwater is not expected at proposed improvement locations. Fluctuations in the groundwater level may occur due to variations in rainfall, temperature, site development, nearby creek flows, and other factors not evident during the field investigation.

5. Geotechnical Observations, Conclusions, and Design Recommendations

The following sections are included primarily for the engineer performing design. If additional geotechnical design recommendations are necessary, GEI should be contacted to provide the required information.

5.1. RSPH Improvements

Several improvements to the RSPH are proposed. The following sections present relevant geotechnical observations and design recommendations for the various improvements.

5.1.1. Power Pole Replacement

The park is planning to replace the existing overhead electric power poles located between the North Rim and the RSPH. Power poles are currently located at a total of four locations along the canyon wall including the North Rim, Upper Bench, Lower Bench, and RSPH. The new power poles will be located generally at the existing power pole locations. Power poles will generally be located on steep, rugged slopes and construction will be difficult. GEI recommends the power poles are not located at the bottom of any gully feature where there is an inherently greater risk of erosion, debris flow, or a rockfall impact.

Based on aerial imagery, geologic maps, and observations from the site visit, the power poles will be located in areas with shallow shale, siltstone, and/or sandstone bedrock underlying a variable thickness of gravelly colluvium or talus. Ideally the selected power pole location will have a minimal soil cover (colluvium or talus) thickness over the bedrock.

GEI understands the power pole foundations will likely consist of a micropile system based on conversations with HDR. Recommendations for the design of micropiles embedded in fresh to moderately weathered bedrock are presented below.

1. Micropiles should be designed and installed in accordance with Case I, Type A1 requirements as specified in U.S.D.O.T publication number FHWA NHI-05-039. Assuming groundwater is not encountered during micropile drilling, GEI anticipates Type A1 micropiles using gravity grouting methods are acceptable. Temporary casing will be needed to maintain hole stability in surficial soils and weathered bedrock.
2. Drilling methods shall be determined by the contractor.
3. Micropiles should have a minimum diameter of 4 inches. Larger diameters may be used, but the area of reinforcing steel required for uplift tension should be increased proportionately with the diameter of the micropile.
4. Micropiles should be designed using grout-to-ground interface bond strength of 25 pounds per square inch (psi) for that portion of the pile in fresh to moderately weathered bedrock and below the top 5 feet of pile length. Grout to ground strength should be neglected for the portion of the micropile extending through soils (colluvium and/or talus anticipated) overlying the bedrock.

Assuming pre-production verification testing and production pile proof testing is performed, a factor of safety of 2 should be used. End bearing pressure should be ignored for micropile design.

5. Micropiles should have a minimum penetration into the bedrock of 15 feet. Longer micropiles may be required for larger loads and when micropiles will be drilled through soil (colluvium or talus) prior to encountering bedrock.

6. By adhering to these recommendations, GEI estimates that foundation settlements should not exceed approximately 1/2 inch.

7. Installation of micropiles should be observed by a representative of the geotechnical engineer to confirm the depth, vertical alignment, the bearing strata, reinforcing steel placement, and grout placement. Grout used in piles should be sampled and tested for strength.

8. Verification load testing should be performed on one or more sacrificial pre-production test piles near locations where the micropiles are planned. Load testing should be performed prior to installation of the production piles in general accordance with the verification load testing procedures described in FHWA NHI-05-039. A representative of the geotechnical engineer should be present during verification load testing to observe and document testing conditions and results. Based on the findings of the verification load tests, design axial capacities may be reevaluated for planned production piles.

9. Proof testing should be performed on production piles in general accordance with the proof testing procedures described in FHWA NHI-05-039. Preliminarily, two or more production piles should be planned for proof testing to account for different micropile orientations with respect to the nearly horizontal shale bedrock bedding orientation, the results of which should be used to assess the need to alter the design pile inclination. Based on the results of the initial proof testing, additional testing may be recommended. The proof testing should be monitored in the field by a representative of the geotechnical engineer. The design engineer should evaluate the need for additional micropiles based on the results of the proof testing program.

10. Micropile groups achieve a net reduction of vertical and lateral capacity with respect to the sum of the individual micropiles. Micropile spacing of greater than 3 micropile diameters center-to-center do not require vertical capacity reduction, and micropile spacing of greater than 5 micropile diameters do not require lateral capacity reduction. Table 5-1 provides group reduction factors for vertical capacity of pile groups, and Table 5-2 provides group reduction factors (also called p-multipliers) for lateral capacity of pile groups.

Table 5-1. Micropile Vertical Group Reduction Factors

Micropile Configuration	Micropile Center to Center Spacing[1]	Reduction Factor
Single Row	2D	0.9
	3D	1.0
Multiple Rows	2.5D	0.67
	3D	0.8
	4D and Greater	1.0

Notes:
1. D refers to the micropile diameter.

Table 5-2. Micropile Lateral Group Reduction Factors

Micropile Center to Center Spacing[1]	Reduction Factors (P-Multiplier)		
	First Row	Second Row	Third Row and Greater
3D	0.8	0.4	0.3
5D	1.0	0.85	0.7

Notes:
1. D refers to the micropile diameter.

11. We understand LPile or a similar modeling program will be used for analysis of lateral loading on the micropiles. For LPile modeling, we recommend using the input parameters listed in Table 5-3 below for micropile lateral analysis. The depth of each material in the model is dependent on the location of the proposed micropile.

Table 5-3. Material Input Parameters for LPile Lateral Analysis

Material	p-y Curve	Effective Unit Weight (pcf)	Friction Angle (deg)	Modulus of Subgrade Reaction k (pci)	Uniaxial Compressive Strength, q_u (psi)	Initial Rock Mass Modulus, E_{ir} (psi)	Rock Strain Factor, k_{rm}	RQD (%)
Colluvium or Talus	Sand (Reese)	130	31	60	-	-	-	-
Bedrock	Weak Rock (Reese)	130	-	-	25	$1x10^5$	0.00005	50

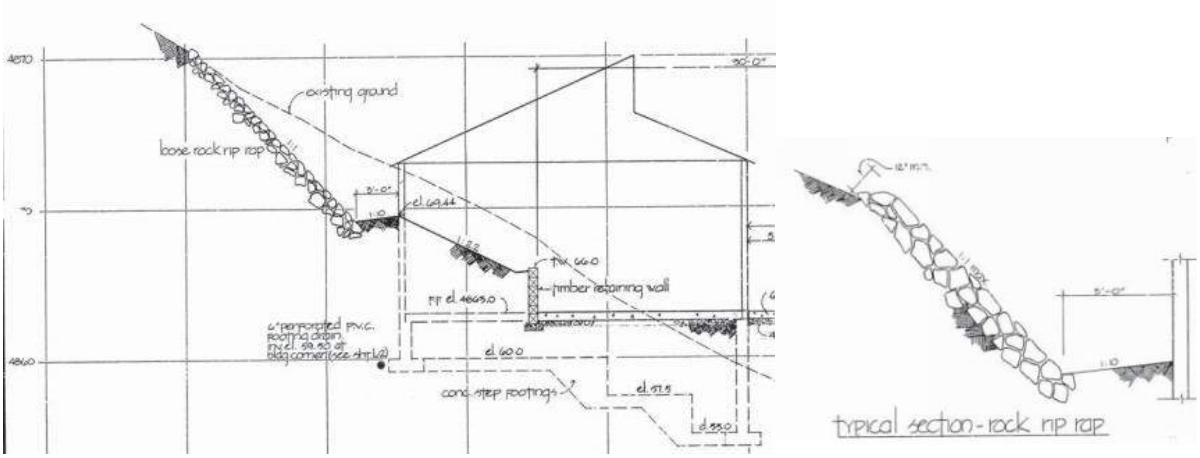
5.1.2. Riprap Slope and Rockfall Protection Behind RSPH

5.1.2.1. Riprap Slope Observations

The Roaring Springs Pumphouse Facility is located on a steep hillside above Roaring Springs Creek. A portion of the slope, uphill and directly southwest of the building and outside piping, is covered with loose riprap. The slope above the pumphouse is variable with grades ranging from approximately 1:1 (Horizontal:Vertical) in the area covered by riprap to 2:1 in areas that do not appear to have been altered during construction of the pumphouse. In general, the area is vegetated with native grasses, cacti, and various size shrubs and trees. The area directly above the existing riprap slope includes a rockfall protection system which is described later in this section. Photos 9-13 in Appendix A depict the riprap slope and rockfall protection system.

The riprap slope appears to have been designed and constructed to reduce erosion caused by the grading that was required to construct the pumphouse. Construction drawings provided by the NPS, dated 8/4/1977 indicate the riprap was constructed at a 1:1 slope with 3 feet between the toe of the riprap slope and the backwall of the pumphouse. The size of individual rocks contained in the riprap were not defined, but a minimum thickness of 12 inches was specified for the riprap slope armoring. Figure 5-1 below presents typical sections of the riprap from the construction drawings referenced above.

Figure 5-1. Typical Riprap Sections



In general, our observations at the site indicated the riprap slope was constructed per plan. The riprap was comprised of mostly angular, cobble to boulder sized limestone and dolomite rock. Typical rock sizes ranged from approximately 4 inches to 24 inches in the longest dimension. Several larger boulders were observed with sizes ranging from 3 to 4 feet in the longest dimension. The majority of the slope was covered with riprap, although several areas were observed where the riprap had slid downhill and exposed the subgrade. Photos 9 and 10 in Appendix A are representative of our observations of the typical riprap adjacent to the pumphouse and areas lacking coverage. We recommend areas of this slope lacking rip rap coverage receive replacement rip rap for coverage in general accordance with the construction drawings.

5.1.2.2. Riprap Slope Conclusions and Recommendations

The riprap is currently pushing against the piping adjacent to the pumphouse. Photos 9 through 12 in Appendix A show the general condition of the pipes and riprap observed during the site visit. If not mitigated, the riprap will continue to fail and impact the piping. Failure of the riprap slope is the result of the overall steepness of the slope, the minimal thickness of the riprap armoring, lack of bedding for the riprap, and the absence of a buttress system at the toe of the riprap slope. Several alternatives exist to mitigate further failure of the riprap, ranging from complete reconstruction of the slope to simple removal of the riprap adjacent to the piping. Complete restoration of the riprap-armored slope would be difficult, expensive, and have significant impacts to the site. Removal of the riprap adjacent to the piping would create an ongoing maintenance effort that would ultimately result in the slope no longer being armored, creating significant erosion related issues.

Based on our discussions with HDR and NPS personnel, we recommend the riprap slope be stabilized using a pinned mesh system consisting of lashed cables or integrated cable mesh such as those manufactured by Geobrugg (Spider S3/S4, Quarox), Maccafferi (MacArmour & SteelGrid HR, HEA Panels), or approved equivalent. Mesh opening size will depend on the size of the riprap being retained. Mesh should be pinned at a spacing of approximately 6-foot by 6-foot to 10-foot by 10-foot on center using pins consisting of steel rebar anchored and cemented into competent bedrock. Pin spacing, length, diameter, and embedment should be verified by performing site-specific analysis. The pinned mesh

system should be used in all existing riprap locations. Care should be taken during demolition and construction to avoid dislodging riprap that could fall and impact the RSPH structure.

5.1.2.3. Rockfall Protection System Observations, Conclusions, and Recommendations

As part of the November 17, 2020 site visit, GEI observed the rockfall protection system constructed upslope of the riprap covered slope (As Constructed Drawings, April 19, 1988). The system includes an approximately 80-foot-long by 8-foot-high wire rope and steel mesh rockfall fence that is anchored into the slope by a series of steel columns and steel cable anchors. The overall condition of the rockfall protection system was good. The fence, columns, and anchor cables appear to be in good shape and there is no evidence of damage caused by rockfall. Several boulders were present on the uphill side of the fence, most likely the result of small rockfall events from a rock outcrop upslope of the system. During the site visit, there was evidence of a relatively small but recent rockfall event. The boulder-sized rock came to rest on the slope and did not reach the fence. Photo 13 in Appendix A shows the general condition of the rockfall protection system observed during the site visit.

Based on the condition and functionality of the existing rockfall protection system, no changes are recommended to the system. The system is in good condition and functioning as designed. However, the NPS should remove any large buildup of materials on the upslope side of the fence to allow for proper function of the system in the future. The captured rockfall materials removed from behind the fence can be disposed of at the site or utilized as additional riprap materials for the slope below. The frequency of material removal is hard to determine without understanding how long it has been since this work was last conducted. The NPS should conduct annual visual assessments of the system to determine if the conditions have changed.

5.1.3. New Fire Water Tank

A new 5,000-gallon fire water tank is proposed on the south side of the RSPH structure (Photo 14, Appendix A). Based on site observations and review of the crib wall and RSPH foundation wall geometry relative to the proposed tank footprint, surcharge loading on the existing crib wall and foundation wall from the proposed tank will be minimized at the proposed location. Therefore, in our opinion this is an acceptable location for the proposed tank.

Subgrade work will be necessary to create uniform bearing conditions. We recommend a minimum 1-foot deep overexcavation occurs below the tank footprint. At the time of overexcavation the geotechnical engineer should observe the exposed soil materials to determine whether voids are present below the overexcavation. If voids are present below the overexcavation, they will need to be backfilled, additional excavation may be necessary to backfill any voids. Following the overexcavation, an AASHTO Class 2 non-woven separation fabric should be placed over the bottom of the overexcavation surface and excavation sidewalls.

The overexcavation should be backfilled with processed, moisture conditioned, and compacted onsite granular fill material in accordance with Section 6.2.4. The onsite material placed at this location should be processed to ensure a uniform gradation with a maximum particle size of 3 inches, 30-70 percent passing the No. 4 sieve, and a maximum of 25 percent passing the No. 200 sieve.

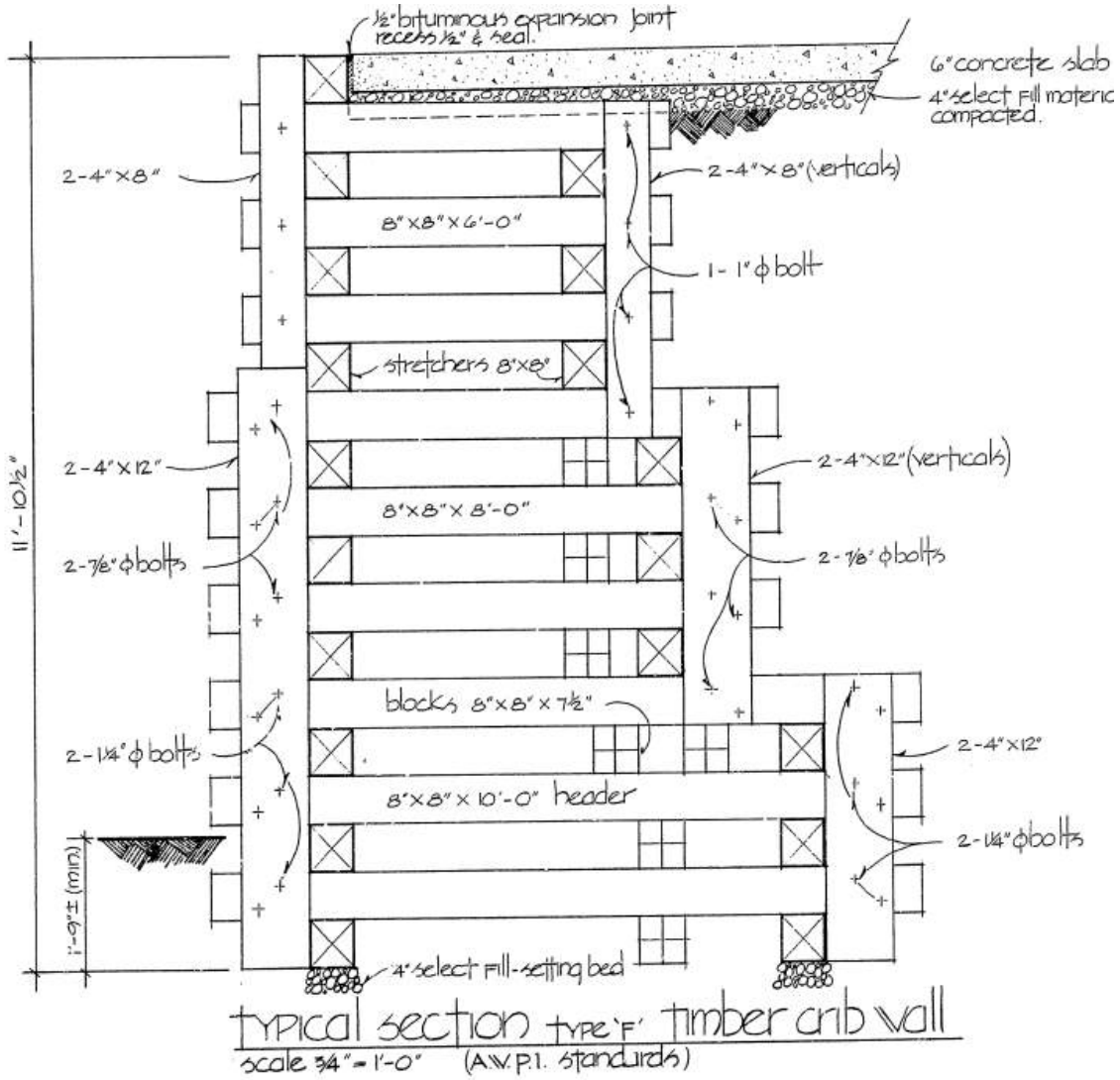
GEI recommends the following criteria be utilized in design of mat or ringwall foundation elements for the proposed tank:

1. The foundations should be designed using a maximum allowable bearing pressure of 2,000 pounds per square foot (psf) This value includes appropriate allowable-stress-design (ASD) safety factors.
2. The foundations should be designed to withstand a total settlement of 1 inch and differential settlements of 0.5 inches.
3. It is acceptable to increase the net allowable bearing capacity by one-third if the load combination utilized considers infrequent loads such as wind or earthquake. Snow loading is not considered an infrequent load condition.
4. Sliding friction between anticipated subgrade and foundation materials should be evaluated using a coefficient of 0.4.
5. The coefficient of subgrade reaction (k) is typically used for flexible foundation analysis and describes the load intensity per unit of displacement. For compacted onsite granular material described above, a k value of 200 psi/inch is recommended.
6. If the tank is located close to the existing RSPH structure and imposes loading on any existing below grade walls, the added load imposed on the existing structure should be evaluated.

5.1.4. Timber Crib Retaining Wall and Concrete Slab

GEI observed the existing timber crib retaining wall on the northeast side of the pumphouse. Photos 15-18 of Appendix A show the general condition of the timber crib retaining wall and concrete slab. The approximately 12-foot-tall retaining wall is supporting the fill that underlies the exposed aggregate concrete slab/sling pad. Based on observations and provided construction drawings, dated 8/4/1977 the retaining wall is constructed with three tiers comprised of horizontal 8x8 rough sawn lumber. The vertical members of the two lower tiers are comprised of two 4x12 rough sawn lumber, while the vertical members on the upper most tier are comprised of two 4X8 rough sawn lumber. The exposed-aggregate concrete slab supported by the retaining wall is 6 inches thick and was designed to have 4 inches of select, compacted fill under the concrete overlying loosely placed fill. The wall appears to be constructed in general accordance with the construction drawings. Figure 5-2 below presents the typical section of the timber crib retaining wall from the construction drawings referenced above.

Figure 5-2. Typical Timber Crib Retaining Wall Section



Given its age and nature of the structure, the overall condition of the timber crib retaining wall is good. Some damage to the rough sawn lumber is visible but does not appear to pose a significant threat to the integrity of the overall structure.

The fill retained by the wall is showing significant signs of degradation. In numerous places the fill is being washed out or eroded between the timbers, resulting in voids behind the wall facing and under the concrete slab. The surface of the slab is in relatively good condition given its age, use, and condition of the fill supporting it. The joints in the concrete slab and connections to the pumphouse and timber crib wall show signs of settlement of the slab and caulking at the joints is nonexistent at some locations and in poor condition at most locations. Poor drainage on the slab appears to be a significant contributing factor to the degradation and erosion of the fill underlying the slab. Overall, the condition of the slab and underlying fill is poor and repairs are recommended to increase the longevity of the structure.

The timber crib retaining wall is in relatively good condition given its vintage and materials utilized in its construction. The majority of the timber members appear to be in relatively good condition. Only a few timbers appear to be splitting. The biggest issue identified by GEI during the site visit was the erosion of the fill underlying the exposed-aggregate concrete slab. The loss of fill materials is resulting in settlement of the concrete slab. As the slab settles, it appears to be creating larger separations along joints and where it connects to the pump house and the outside edge of the timber crib wall. With time, water infiltration from precipitation and potentially other sources related to operations and maintenance of the pumps will continue to erode the fill underlying the slab. If this continues, the slab will continue to settle and exacerbate the issue.

To limit the potential for further void development below the slab, some or all of the crib wall height should be permanently faced with wire mesh or wooden slats. The facing should extend from the top of the crib wall to the bottom of the crib wall. Aggregate consisting of 1/4-inch to 3/4-inch size rock should be placed behind the selected facing material where voids currently exist using vibratory methods to achieve a suitable relative density. If wooden slat facing is selected, AASHTO Class 2 non-woven separation fabric should be placed between the crib wall and the wooden facing. If wire mesh is selected, the mesh should be sized such that openings are smaller than the minimum particle size of rock placed behind the wire mesh. Facing should be designed such that moisture behind the facing is allowed to freely flow through the facing to minimize the presence of long-term moisture and subsequent wood rot.

If any of the concrete slab will remain, we recommend the NPS consider caulking all the joints between the existing concrete panels, joints between the pumphouse and the slab, and joints between the top timbers and the slab. In addition, the NPS should complete an annual inspection of the caulking and re-caulk areas of missing or weathered caulking. The caulking will help slow the erosion of the fill but will likely not be 100 percent effective, especially as large portions of the slab are not covered with concrete.

Where concrete slab replacement is proposed, AASHTO Class 2 non-woven separation fabric should be placed below new subgrade to limit migration of fine soil particles and further development of voids below the slab. We observed many areas along the crib wall where about 1 foot of soil is missing below the slab and this subgrade will need to be replaced. The new concrete slab should be supported on a minimum 6-inch-thick zone of processed, moisture conditioned, and compacted onsite granular fill material in accordance with Section 6.2.4. The onsite granular fill material to be placed at this location should have a maximum particle size of 3 inches, 30-70 percent passing the No. 4 sieve, and a maximum of 25 percent passing the No. 200 sieve.

The timber members exhibiting distress should be repaired or replaced. Replacement of distressed vertical members will be difficult and require significant shoring of the system. Rather than replacement, it may be possible to add steel bands around the split timbers. Distressed horizontal members, specifically at the top of the wall adjacent to the concrete slab, should be relatively easy to replace.

5.2. RSPH Site to Manzanita Ranger Station Power Distribution

GEI walked along the existing Trans-Canyon Waterline (TCWL) alignment to observe the ground conditions where a buried power line is proposed connecting RSPH and Manzanita (Photos 6 & 8, Appendix A). During the site visit, GEI observed that the existing pipe is exposed on the surface or very

thinly covered for most of the alignment. Running new power cables along this alignment is feasible. In most cases, it will be easiest to strap the cables to the TCWL. In other areas it can be buried in a separate trench or strapped to the rock outcrops.

Burying the TCWL and power cables below existing grade along this reach will be very challenging. Borrow sources for cover soils are very limited near the alignment and often consist of large cobbles and boulders, which will generally perform marginally as cover materials. In many places the TCWL sits on hard sandstone bedrock or is tucked into overhanging rock ledges making burial difficult (Photo 8, Appendix A).

Based on conversations with HDR, we recommend the power cables be suspended from steel pipe supports with a maximum spacing of 10 feet where strapping the cable to the TCWL or burying the cable is not feasible. Specifically, we understand that approximately 50 feet of the cable between stations 1120+00 and 1121+00 will require pipe supports. We recommend the pipe supports be anchored to the exposed rock using steel anchor bolts. Bolt holes should be drilled a minimum of 12 inches into competent rock and grouted using a cementitious grout following bolt placement. Individual bolts should be designed for a maximum pullout strength of 6 kips.

5.3. Cottonwood Campground Site Power Distribution

We understand two alternate locations are being considered for a proposed photovoltaic shade canopy structure at Cottonwood Campground. Based on the findings of the test pit exploration near each alternate location, we anticipate the soils at each location are similar and will consist of alluvial silty sand with gravel and cobbles as shown in Photos 3 and 4 in Appendix A. We observed a limited extent of fill material with wood debris on the northeast side of the existing storage building. If existing fill is encountered within the proposed structure excavation footprint during construction, the existing fill should be removed to expose native soil and be replaced, moisture conditioned, and compacted in accordance with Section 6.2.4.

GEI recommends the following criteria be utilized in design of shallow pad or column foundation elements for the proposed canopy structure bearing on firm sandy or gravelly alluvium:

1. The foundations should be designed using a maximum allowable bearing pressure of 2,000 psf This value includes appropriate allowable-stress-design (ASD) safety factors.
2. The foundations should be designed to withstand a total settlement of 1 inch and differential settlements of 0.5 inches.
3. It is acceptable to increase the net allowable bearing capacity by one-third if the load combination utilized considers infrequent loads such as wind or earthquake. Snow loading is not considered an infrequent load condition.
4. Sliding friction between anticipated subgrade and foundation materials should be evaluated using a coefficient of 0.35.
5. The coefficient of subgrade reaction (k) is typically used for flexible foundation analysis and describes the load intensity per unit of displacement. For native sandy or gravelly alluvium

compacted and moisture conditioned in accordance with Section 6.2.4, a k value of 125 psi/inch is recommended.

5.4. Lateral Earth Pressures

Lateral earth pressures presented in Table 5-4 and Table 5-5 below should be considered during the structural design process for foundation elements extending below grade. Lateral earth pressure values are a function of the properties and the geometry of the retained fill, soil, and/or bedrock and anticipated magnitude of lateral deflection. Groundwater was not encountered within the geotechnical investigation, therefore, additional loads due to groundwater were not considered in the lateral earth pressures.

Additional lateral earth pressure due to seismic loading was assessed using a site modified peak ground acceleration of 0.23g as determined using the ASCE 7 hazard tool with inputs consisting of the ASCE 7-22 Standard, Site Class D, and a Structure Risk Category of III.

Table 5-4. Static Lateral Earth Pressures

Backfill Material Type	Unit Weight (pcf)	Friction Angle (degrees)	Static Earth Pressure Coefficients			Static Equivalent Fluid Pressure (pcf)		
			Active	At-Rest	Passive	Active	At-Rest	Passive[1]
Onsite Granular Fill[2]	130	31	0.32	0.48	3.12	42	63	203

Notes:
1. Values shown are half of theoretical calculated value due to large strains required to mobilize passive earth pressure.
2. Granular Fill shall consist of sand or gravel when classified in accordance with USCS criteria.

Table 5-5. Additional Lateral Earth Pressures due to Seismic Loading

Backfill Material Type	Point of Application of Additional Loading[3]	Seismic Earth Pressure Coefficients			Static Equivalent Fluid Pressure (pcf)		
		Active	At-Rest	Passive	Active	At-Rest	Passive[1]
Onsite Granular Fill[2]	0.6H	0.07	0.10	-0.18	9	14	-12

Notes:
1. Values shown are half of theoretical calculated value due to large strains required to mobilize passive earth pressure.
2. Granular Fill shall consist of sand or gravel when classified in accordance with USCS criteria.
3. As measured from the base of the wall where H = total height wall height.

For the lateral earth pressures presented above to remain applicable, backfill material placement adjacent to below-grade walls should be in accordance with procedures outlined in this report. In addition, backfilled material must be placed within a one horizontal to one vertical (1H:1V) backfill geometry, up and away from the base of the structure. Consideration has not been given to vertical loads applied to the backfill surfaces during or after construction as a result of traffic and/or other surcharge loads.

5.5. Frost Protection

In colder climates, certain soils can freeze and can cause heave resulting in larger than expected differential movement. Typically, clays and clean sands or gravels have negligible frost susceptibility whereas silts, silty sands, and very fine sand have high frost susceptibility. Additionally, as moisture contents increase so does the potential for frost heave. We recommend that perimeter shallow foundations and shallow foundations in non-heated enclosed areas have a minimum cover of 24 inches for frost protection per Coconino County Building Codes, Ordinances, and Design Criteria. Frost depth recommendations for buried pipelines should be provided by the pipeline designer.

5.6. Seismic Site Classification

Based on the International Building Code and the American Society of Civil Engineers Minimum Design Loads for Buildings and other Structures, our subsurface investigation, and our experience and knowledge of seismic conditions in the area, a Site Classification of B is recommended for seismic design at the RSPH and Cottonwood Campground sites.

6. General Construction Considerations

The following sections are intended for the Engineer producing specifications and the contractor constructing the proposed project. Construction considerations include temporary excavations, site grading and earthwork, bedrock or oversize material excavation, construction dewatering, and unstable subgrade conditions.

6.1. Temporary Excavations

General site safety including temporary excavations are the sole responsibility of the contractor performing construction. GEI is providing temporary excavation information strictly as an informational benefit to the project team, specifically the general contractor. An Occupational Safety and Health Administration (OSHA) defined competent person should be identified by the contractor to be in charge of temporary excavations. In general, the contractor's competent person should have experience or training in determining soil types, benching and shoring, and have the ability to detect potential temporary slope stability and protective system issues.

OHSA defines an excavation as a man-made cut, trench, or depression formed by the removal of earth. A trench is a specific type of narrow excavation with a geometry including a greater depth than width and a width of 15 feet or less. Trenches 5 feet deep or greater should be sloped, retained with shoring, or shielded appropriately. Shielding most commonly includes trench boxes. In general, shoring can include inclined, horizontal, or vertical systems depending on the excavation geometry and availability of retention alternatives. Sloping and benching should be in accordance with OSHA recommendations. Benching should include a maximum 4-foot vertical face for each bench and the overall excavation geometry less than or equal to the OSHA defined slope.

If deep excavations are proposed, a professional engineer registered in the State of Arizona should approve the contractor's approach for trenches greater than or equal to 20 feet in depth. In addition, shoring for trench excavations should be designed by a professional engineer or be based on tabulated data prepared or approved by a registered professional engineer in accordance with OSHA 29 CFR 1926.652(b) and (c).

GEI has evaluated observed soil conditions likely to be penetrated by the proposed construction. Based on the OSHA determined soil Types, Table 6-1 below presents maximum recommended temporary excavation slopes to be utilized during construction.

Table 6-1. OSHA Temporary Excavation Slopes

Backfill Material Type	OSHA Classification	Maximum Recommended Slope (H:V)[1,2]
Alluvium and Colluvium	Type C	1 1/2:1
Bedrock	Type A	3/4:1

Notes:
1. H:V is an abbreviation for Horizontal:Vertical.
2. Valid for trench excavations less than 20 feet in depth.

During construction, heavy equipment or excavated material stockpiles should be kept away from excavation edges to the extent possible. It is recommended that heavy equipment, construction material stockpiles, or excavated material stockpiles be kept at least 10 feet from the edges of excavations. Existing underground utilities should be fully understood and documented prior to initiating excavations including information regarding location, depth, type, and size. Finally, the contractor's competent person should inspect trenches and excavations routinely for signs of instability including sliding, toppling, subsidence and bulging, heaving or squeezing, boiling, and/or other visual concerns.

6.1.1. Shored Temporary Excavations

The contractor may elect to use a temporary shoring system to limit excavation extents, particularly if other site infrastructure may be impacted. For excavations greater than 20 feet deep, the contractor will be required to submit a design prepared by a professional engineer registered in the state of Arizona to demonstrate that the shoring system maintains both local and global slope stability for the duration of the open excavation.

6.2. Site Grading and Earthwork

Appropriate site preparation, material placement and compaction, and backfill selection can reduce the risk of post-construction settlement and potential issues related to lateral earth pressures. General site grading and earthwork considerations are presented in the following sections.

6.2.1. Site Preparation

Areas supporting backfill and pipe bedding should be properly prepared. Once the rough grade has been established in excavated areas, and prior to placement of backfill, the exposed subgrade should be carefully inspected via probing and testing, as determined necessary by the geotechnical engineer and as outlined in Section 6.2.4. At the time of such observation, it may be necessary to make hand auger borings and/or use a hand penetration device in the base of the foundation excavation to confirm that the soils below the base are satisfactory for foundation support. The necessary depth of penetration will be established during observation. Frozen, wet, soft, or loose soil, as well as any other undesirable material should be removed. Once suitable soil conditions are achieved, the subgrade should be scarified and compacted prior to fill placement as described in Section 6.2.4. The suitable exposed soil materials should be scarified and moistened or dried, as necessary, to a minimum depth of 8 inches below the proposed construction. Scarified material should be compacted to the minimum specifications defined in the following sections.

6.2.2. Structure and Trench Backfill Material

Backfill is anticipated below foundations, adjacent to structure walls, and within pipeline trenches. Please refer to Section 5.1 for specific fill material recommendations at the RSPH. Other pipeline trenches and foundation walls should be backfilled with onsite granular fill materials having more than 50 percent retained on the No. 200 sieve (i.e., classifying as sand or gravel in accordance with USCS criteria), and with no particles exceeding 3 inches. Ideally, sandstone or limestone bedrock should not be used as backfill unless it is processed to the gradations noted above. Shale or claystone should not be

used as backfill. If another material is considered for any of the presented applications, GEI should be contacted to review submitted particle size distribution and plasticity testing results.

6.2.3. Pipe Bedding

Piping for the proposed pipelines is anticipated to be within alluvium, colluvium, bedrock. Piping should be bedded in a clean granular material from a depth of 6 inches below the pipe to 6 inches above the pipe. Bedding can consist of ASTM C33 No. 67 coarse aggregate with less than 5 percent. If another onsite material is proposed for pipe bedding, GEI should be notified and may adjust geotechnical recommendations accordingly.

6.2.4. Fill Placement and Compaction

We recommend fill placement occurs in maximum 8-inch loose lifts for fill under and backfill adjacent to structures. Lift thickness for pipe bedding should not exceed 6 inches. Minimum recommended compaction specifications are outlined in Table 6-2 below.

Table 6-2. Fill Placement and Compaction

Backfill Material Type	Moisture and Compaction Specifications		
	Moisture Content[1]	Dry Density[2]	Relative Density[3]
Pipe Bedding	-	-	≥ 65%
Onsite Granular Fill	-2% to +2%	≥ 95%	-

Notes:
1. Moisture content relative to the optimum moisture content as determined by ASTM D698.
2. Dry density relative to the maximum dry density as determined by ASTM D698.
3. Relative Density as determined by ASTM D4263 and D4254.

Mechanical compaction is required for all materials placed as backfill during construction. Compaction of cohesive materials is best accomplished with equipment such as a jumping jack or padfoot roller. Non-cohesive soils are best compacted with a vibratory plate or vibratory smooth-drum roller. Compaction utilizing any flooding or jetting type technique should not be allowed. Care should be taken when compacting fill adjacent to structures. Generally, we recommend operating only light-weight compaction equipment such as jumping jacks and vibratory plates immediately adjacent to structures.

Quality assurance of backfill material and backfill placement will be necessary to reduce potential for long term differential settlements. Inspection of subgrade materials prior to placing or forming and casting structural elements is also critical to project success. GEI recommends a qualified testing agency is retained to provide quality assurance services during the backfill process.

6.3. Bedrock or Oversized Material Excavation

Based on our visual field reconnaissance, bedrock is commonly present at the sites beneath a thin layer of alluvium or colluvium. Bedrock excavation is anticipated during proposed construction activities. P-wave velocities measured in the rock within the inner canyon ranged from approximately 13,000 to 14,000 feet/second indicating relatively high intact rock strength. GEI anticipates the Bright Angel Formation and weathered sections of the Tapeats Sandstone can be excavated with conventional

equipment that can be transported into the canyon. This will likely consist of a mini excavator with both a conventional and rock bucket as well as a hydraulic rock breaker (hoe ram). There may be stronger bedrock beds that will likely require specialized equipment. Light blasting and/or a hydraulic rock breaker may be necessary for significant excavations in bedrock. Additionally, specialized equipment may be required to relocate large boulders as needed.

6.4. Construction Dewatering

Groundwater was not observed at the proposed improvement locations. Significant groundwater is not expected at proposed improvement locations during construction. However, fluctuations in the groundwater level may occur due to variations in rainfall, temperature, site development, nearby creek flows, and other factors not evident during the field investigation.

6.5. Unstable Subgrade Conditions

If soft/loose pockets or undesirable materials are encountered in the excavations, the proposed excavations may be re-established by backfilling after the material has been removed. The backfill should consist of coarse granular material such as ADOT Class 1 or 2 base course or similar. If the base of the overexcavation is still soft and further overexcavation is not practical, then a stabilization product meeting the minimum requirements of AASHTO Class 4A Woven Stabilization fabric should be placed in the base of the overexcavation per the manufacturer's recommendations. The backfill should be placed and compacted according to specifications outlined in this report. Field observations by a geotechnical engineer and testing of the open excavations should be performed prior to concrete placement to verify the subgrade conditions and/or to make corrective recommendations, as necessary.

7. Limitations

This study was conducted in accordance with generally accepted geotechnical engineering and engineering geologic practices and principles; no warranty, express or implied is made. The subsurface conditions described in this report were based on data obtained from field observations and minimal hand exploration, information provided by the client, engineering judgement, and our experience with similar subsurface conditions and projects. Subsurface conditions are typically variable, both laterally and vertically, and the nature and extent of the subsurface variations across the site may not become evident until construction. The boundaries between different soil types and bedrock presented in this report are approximate and, in some cases, may be more abrupt or gradational than described herein. Groundwater levels may vary with time, adjacent surface water levels, precipitation, and changes to the hydrogeological conditions at or surrounding the project site.

This report has been prepared exclusively for our client for design purposes for the subject project. GEI is not responsible for technical interpretations by others of the data presented in this report or use of this report by others for the subject project or other projects. If differing site conditions are encountered during further evaluation of the subsurface conditions by others or during construction, GEI should be notified immediately to determine if any changes to our recommendations presented in this report are warranted.

The recommendations presented in this report are only intended for the proposed design and construction as understood by GEI at the time of issuing this report. If the proposed design and construction changes, GEI should be notified immediately and given the opportunity to review the proposed changes and if necessary, modify our recommendations presented herein.

An environmental assessment was not included in GEI's scope of work for this project. Any statements regarding the absence or presence of hazardous and/or toxic substances presented herein are only intended for informational purposes. If the client is concerned about the environmental conditions at the site, GEI recommends the client and/or owner retain a qualified environmental firm to conduct an environmental site assessment.

8. References

American Association of State Highway and Transportation Officials (AASHTO), Standard Specification for Geosynthetic Specification for Highway Applications (AASHTO M 288), Washington, DC.

American Society of Civil Engineers (ASCE) (2022), Minimum Design Loads and Associated Criteria for Buildings and Other Structures (ASCE/SEI 7-22), Reston, VA.

Arizona Department of Transportation (ADOT) (2021), Standard Specifications for Road and Bridge Construction, Phoenix, AZ.

ASTM Standards, ASTM International, West Conshohocken, PA (2012).

Coconino County Arizona, Climatic and Geographic Design Criteria – Coconino County, Flagstaff, AZ (2019).

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Huntoon, P.W., Billingsley, G.H., Sears, J.W., Ilg, B.R., Karlstrom, K.E., Williams, M.L., Hawkins, David (1996), Geologic Map of the eastern part of the Grand Canyon National Park, Arizona.

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USGS (2024), Quaternary Fault and Fold Database of the United States, accessed online at https://earthquake.usgs.gov/cfusion/qfault/show_report_AB_archive.cfm?fault_id=991&section_id= on January 12.

Figures

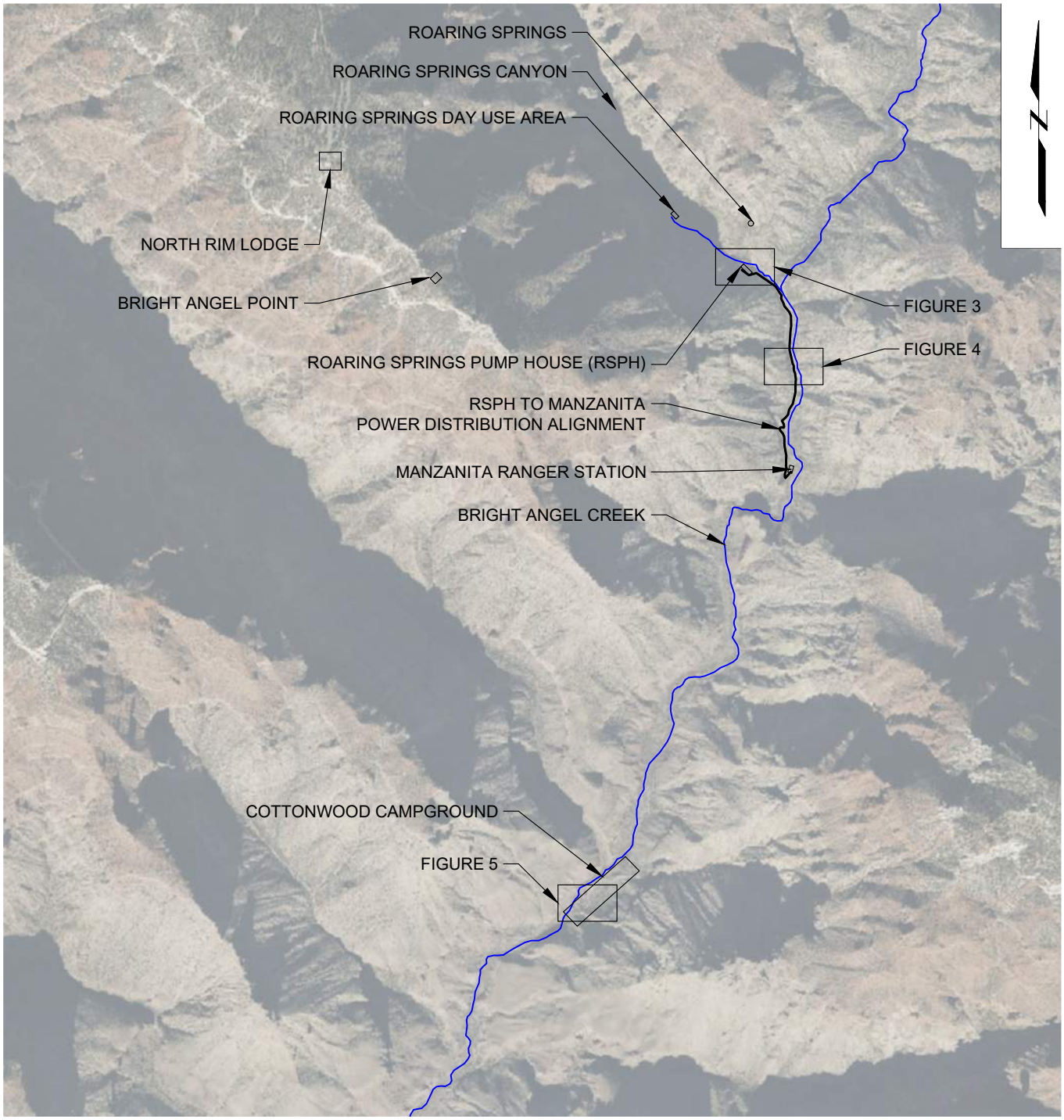
Figure 1. Inner Canyon Site Vicinity Map

Figure 2. Excerpt From Geologic Map

Figure 3. Photo and Test Pit Locations

Figure 4. Photo and Test Pit Locations (Cont.)

Figure 5. Photo and Test Pit Locations (Cont.)

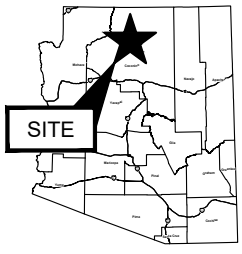


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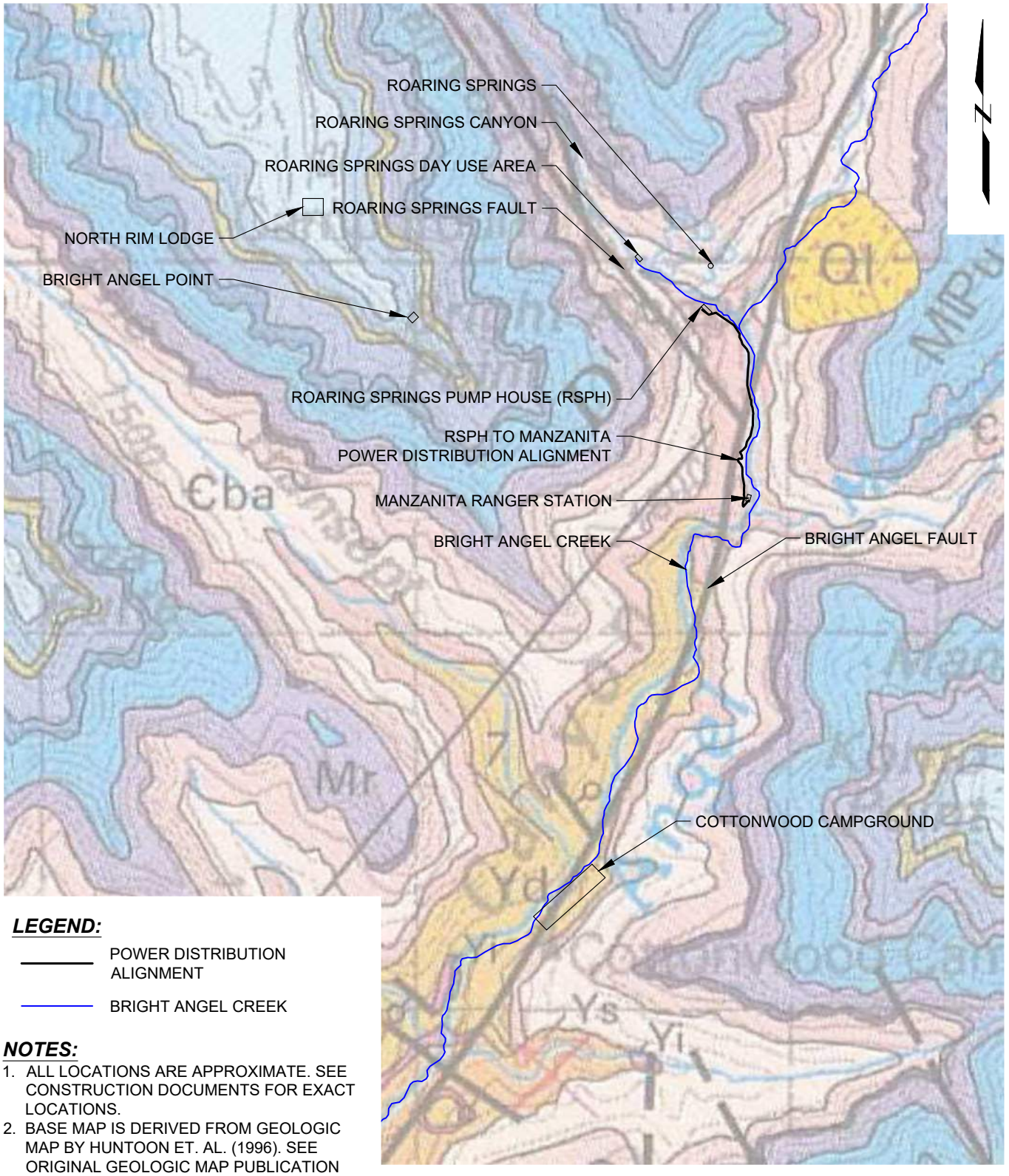
SCALE, FEET



NOTES:

1. ALL LOCATIONS ARE APPROXIMATE. SEE CONSTRUCTION DOCUMENTS FOR EXACT LOCATIONS.

Geotechnical Investigation Report GRCA 318719 Grand Canyon Inner Canyon, Arizona	GEI Consultants	INNER CANYON SITE VICINITY MAP
HDR Engineering, Inc. Fort Collins, Colorado	Project 2305987	August 2025 Fig. 1



LEGEND:

POWER DISTRIBUTION ALIGNMENT

BRIGHT ANGEL CREEK

NOTES:

1. ALL LOCATIONS ARE APPROXIMATE. SEE CONSTRUCTION DOCUMENTS FOR EXACT LOCATIONS.
2. BASE MAP IS DERIVED FROM GEOLOGIC MAP BY HUNTOON ET. AL. (1996). SEE ORIGINAL GEOLOGIC MAP PUBLICATION FOR LEGEND AND UNIT DESCRIPTIONS.

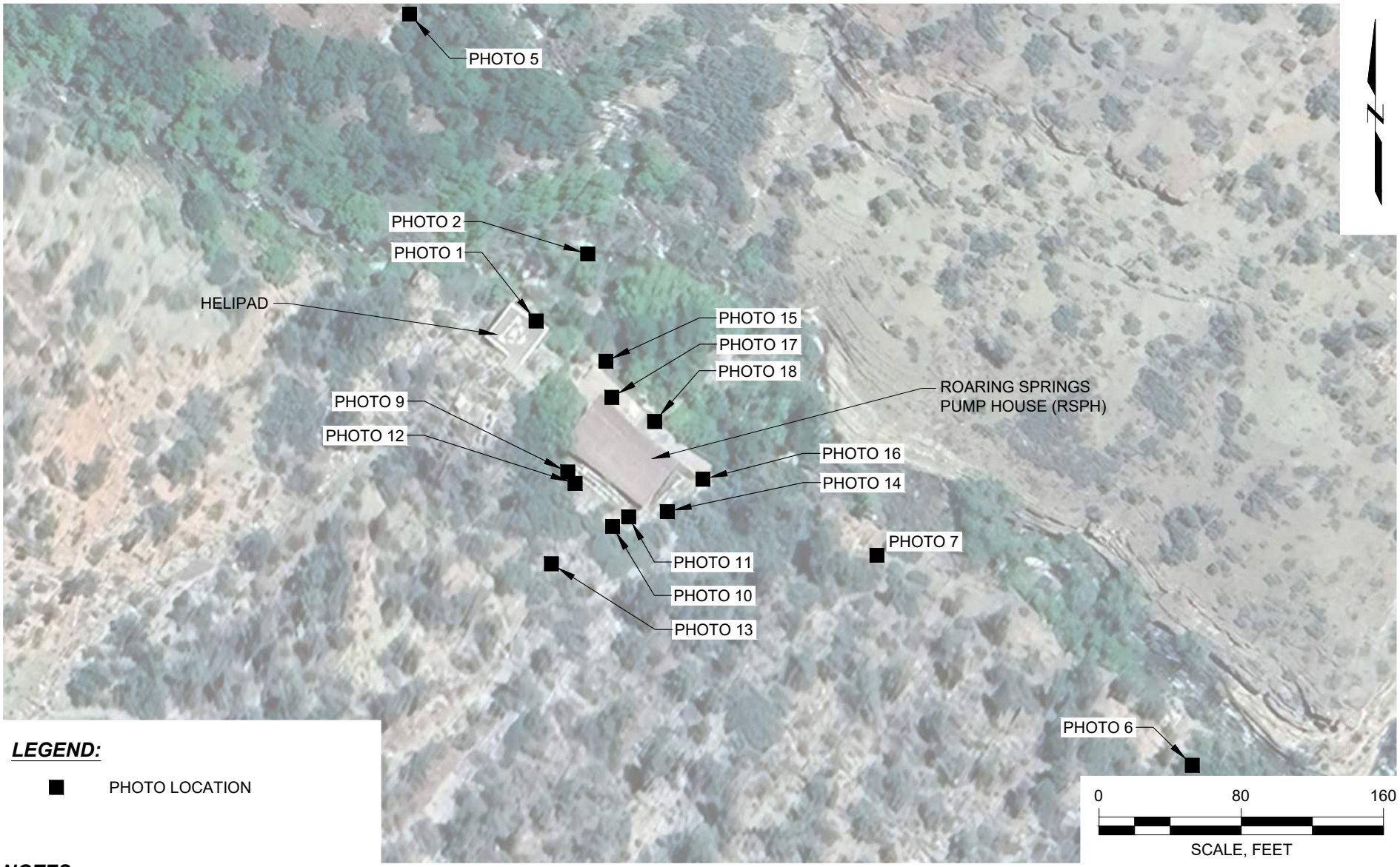
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1. PLAN BASED ON MAP PREPARED BY HUNTOON ET. AL. (1996).

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SCALE, FEET

Geotechnical Investigation Report GRCA 318719 Grand Canyon Inner Canyon, Arizona	GEI Consultants	EXCERPT FROM GEOLOGIC MAP
HDR Engineering, Inc. Fort Collins, Colorado	Project 2305987	August 2025 Fig. 2



LEGEND:

■ PHOTO LOCATION

NOTES:

1. ALL STRUCTURE LOCATIONS ARE APPROXIMATE. SEE CONSTRUCTION DOCUMENTS FOR EXACT LOCATIONS.

Geotechnical Investigation Report
GRCA 318719
Grand Canyon Inner Canyon, Arizona

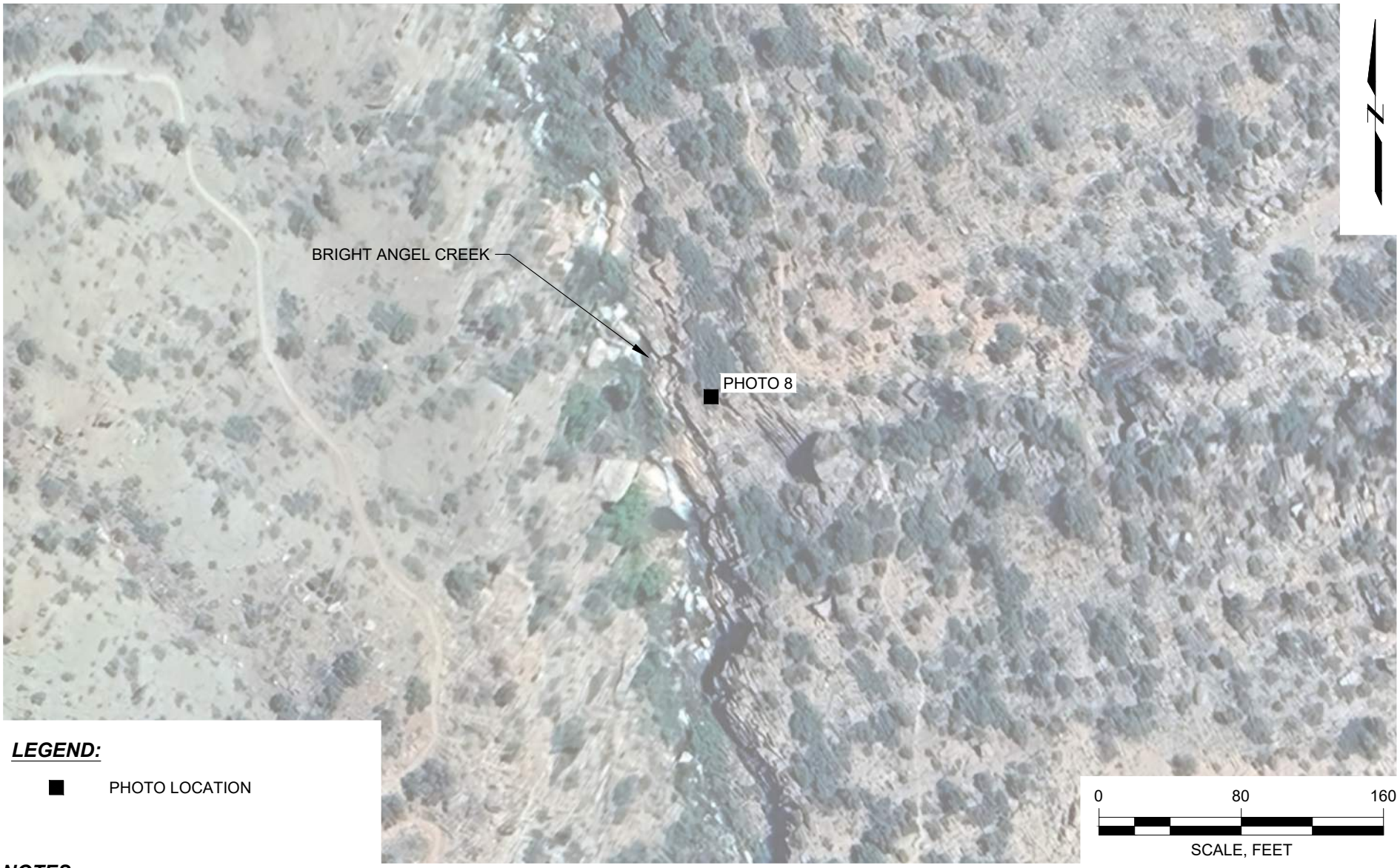
HDR Engineering, Inc.
Fort Collins, Colorado

GEI Consultants

PHOTO AND TEST PIT LOCATIONS

Project 2305987 | August 2025 | Fig. 3

JOHANSON, ROBERT B:\Working\HDR\2305987 21186 GRCA North Rim Water Supply LR\00_CAD\Figures\Inner Canyon GIR\Fig-3.dwg - 5/27/2025



LEGEND:

■ PHOTO LOCATION

NOTES:

1. ALL STRUCTURE LOCATIONS ARE APPROXIMATE. SEE CONSTRUCTION DOCUMENTS FOR EXACT LOCATIONS.

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Grand Canyon Inner Canyon, Arizona

HDR Engineering, Inc.
Fort Collins, Colorado

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PHOTO AND TEST PIT LOCATIONS (CONT.)

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LEGEND:

- PHOTO LOCATION
- TEST PIT LOCATION

NOTES:

1. ALL STRUCTURE LOCATIONS ARE APPROXIMATE. SEE CONSTRUCTION DOCUMENTS FOR EXACT LOCATIONS.

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HDR Engineering, Inc.
Fort Collins, Colorado

GEI Consultants

PHOTO AND TEST PIT LOCATIONS (CONT.)

Project 2305987 | August 2025 | Fig. 5

Appendix A Photographs



Photo 1 – Overview of existing pipe crossing of Roaring Springs Creek below RSPH, viewing downstream from helipad. Alluvial stream terrace with small boulders on right side of stream.



Photo 2 – Pipe crossing of Roaring Springs Creek below RSPH, viewing alluvial stream terrace with small to large boulders on west side of stream.



Photo 3 – Test Pit TP-1, excavated to depth of 8 inches on southwest side of existing storage building at Cottonwood Campground. Soil consisted of loose, dry, silty sand with gravel.



Photo 4 – Test Pit TP-2, excavated to depth of 18 inches on northeast side of existing storage building at Cottonwood Campground. Soil consisted of loose, dry, silty sand with gravel.



Photo 5 – View of RSPH helipad, power poles, and overlying slopes toward the southwest from across canyon. Note gravelly, cobbly, colluvium soil present on slopes where cliffs aren't present, and cliffs consisting of Bright Angel Formation.



Photo 6 – Overview of RSPH to Manzanita TCWL typical partial burial conditions. Note gravelly colluvial soils present.



Photo 7 – Contact between Tapeats Sandstone and overlying Bright Angel Formation observed at bottom of canyon below RSPH (contact represented by dashed yellow line)

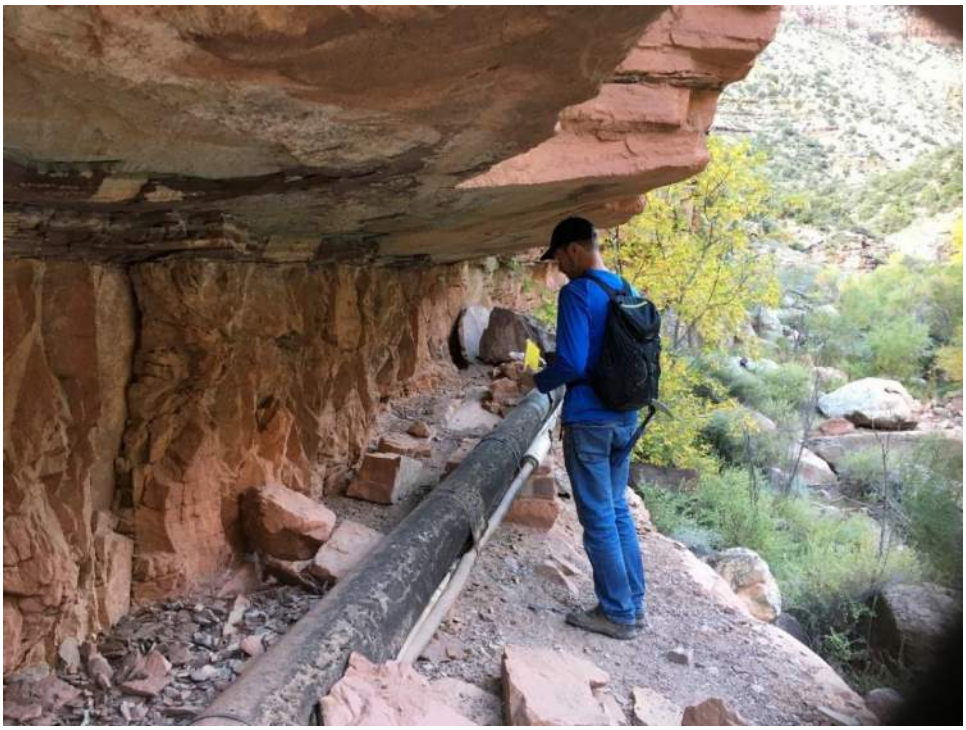


Photo 8 – RSPH to Manzanita TCWL exposed below overhung cliff of Tapeats Sandstone.



Photo 9 – Overview of typical intact riprap behind RSPH structure.

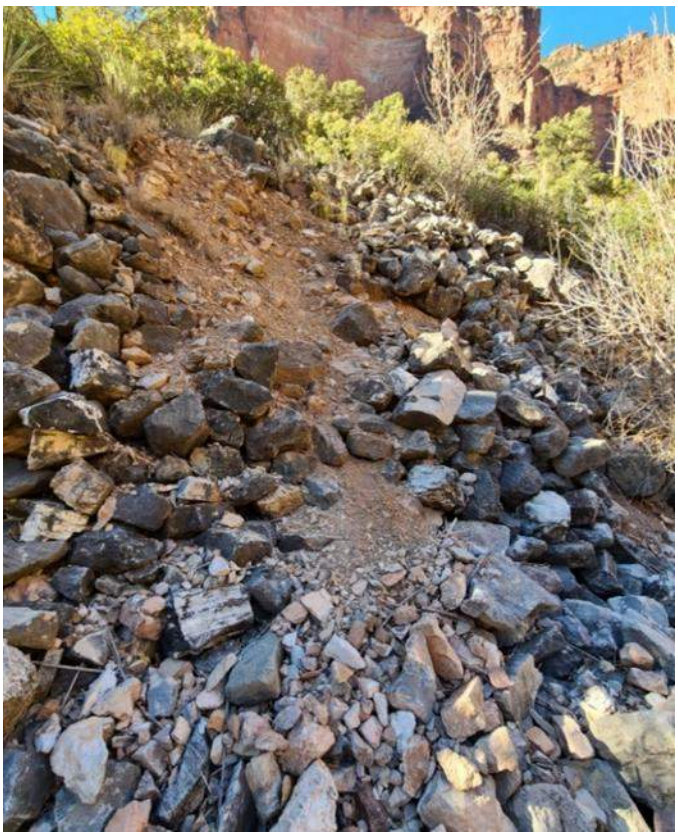


Photo 10 – Overview of failed riprap area behind RSPH structure, with missing riprap exposing brown colluvial subgrade soil.



Photo 11 – Piping adjacent to pumphouse looking northwest. Note riprap in direct contact with outermost pipe.



Photo 12 – Piping adjacent to pumphouse looking southeast. Note riprap in direct contact with outermost pipe.



Photo 13 - Typical section of rockfall protection system. Note steel columns and steel cable anchors.



Photo 14 – Proposed 5,000-gallon fire water tank location at RSPH



Photo 15 – Middle and top tiers of timber crib wall at RSPH. Note cracked and separated vertical member (blue arrow) and loss of backfill material behind timbers typical of existing loose fill.



Photo 16 – Missing backfill material typical of erosion of existing loose fill. Note void and exposed bottom of concrete slab.



Photo 17 – Concrete slab adjacent to pumphouse. Note separation between pumphouse and slab resulting from settlement of the slab and underlying fill. Missing caulking filling joints between concrete panels and between slab and pumphouse allows for water infiltration and results in erosion of underlying fill.



Photo 18 – General condition of timber crib wall and slab. Note portions of slab are unpaved and appear to be compacted fill. Note spot repairs to fill, slab, and split horizontal lumber element.